Dataset Cartography

- **The following paper introduces (2020) Data Maps - a model-based tool to characterize and diagnose datasets:**
- **Swayamdipta, S., Schwartz, R., Lourie, N., Wang, Y., Hajishirzi, H., Smith, N., Choi, Y., Dataset Cartography: Mapping and Diagnosing Datasets with Training Dynamics. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, November 16-20, 2020, pp. 9275-9293.**

- **The source of information used for building data maps: the behavior of the model on *individual instances* during training i.e. *the training dynamics*.**

- **Remark: not all examples contribute equally towards learning.**

- **Two intuitive measures are used for each example:**
  - **the model's confidence in the *true* class (the y – axis)**
  - **the variability of this confidence across epochs – obtained in a single run of training (the x – axis)**

Regions in the Data

The built data maps show the presence of three regions in the data:

- **ambiguous examples regions (with respect to the model; displaying high variability)**
- **easy-to-learn examples regions (low variability, high confidence)**
- **hard-to-learn examples regions (low variability, low confidence)**

- **The working hypothesis (Swayamdipta et al., 2020):**

  <u>Ambiguous</u> and <u>hard-to-learn</u> regions could be the most informative for learning, since these examples are the most challenging for the model!

This is an example of a Data Map:

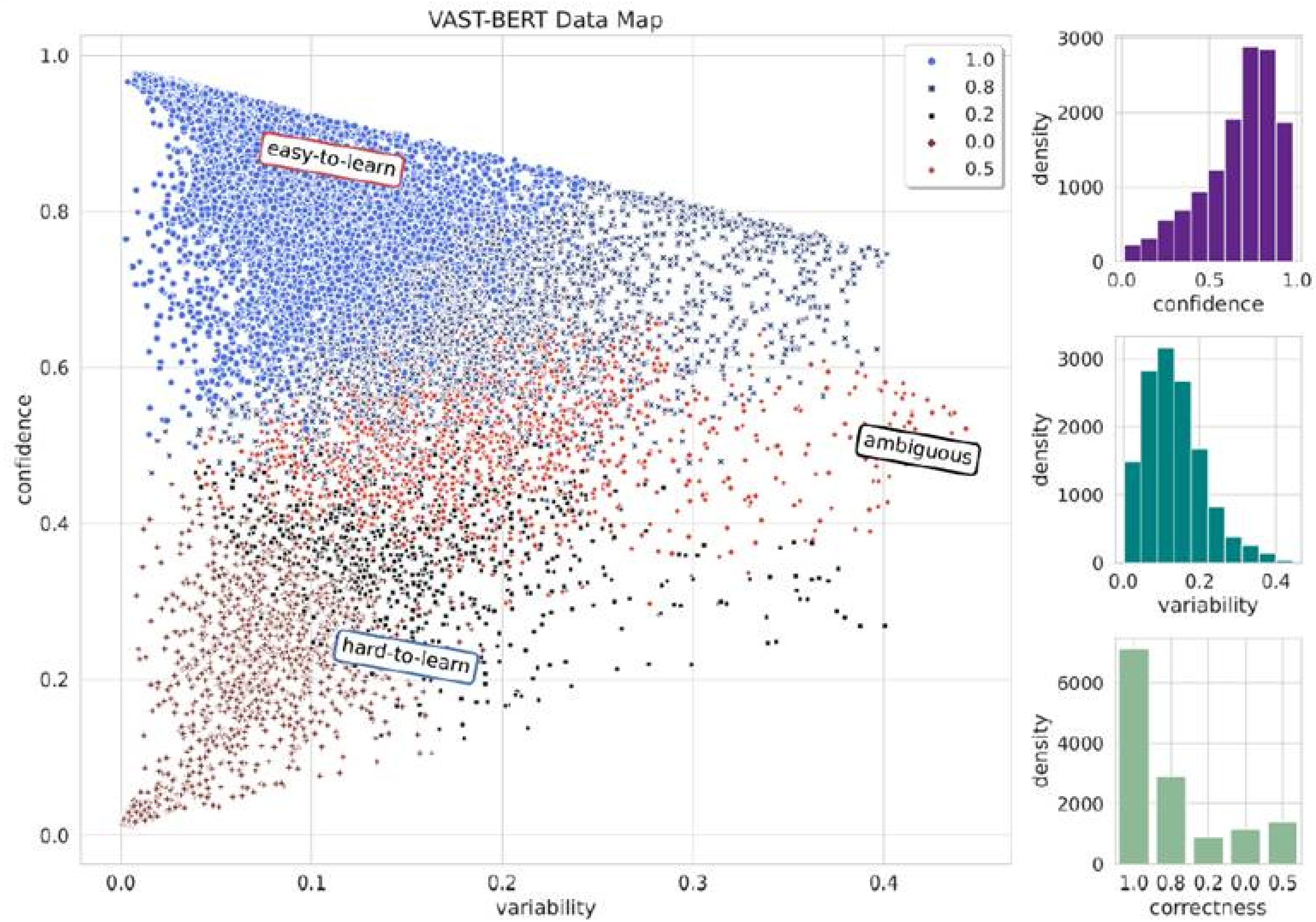

Figure 3.1: Data Map for VAST Dataset, BERT model

Remarks

- Datasets contain a majority of *easy-to-learn* instances; without any such instances training could fail to converge.
- *Easy-to-learn* instances aid optimization.
- *Hard-to-learn* instances frequently correspond to labeling errors.
- *Hard-to-learn* does not perform as well as *ambiguous* on most cases (see test results).
- *Ambiguous* instances alone might be insufficient for learning.
- *Ambiguous* instances are however useful for high performance.

The goal:

- to help visualize a dataset with respect to a model
- to understand the contributions of different groups of instances towards that model's learning

The method:

- the three regions are investigated by training models exclusively on examples from each region

<u>Training Dynamics</u>

- **Consider**

  $$\mathcal{D} = \left\{ \left(x, y^* \right)_i \right\}_{i=1}^{N}$$

  a *training set* of size N, where the i-th instance consists of
  - $x_i \equiv$ **the observation**
  - $y_i^* \equiv$ **its true label under the task**

- **The method assumes a particular model whose parameters are selected to minimize empirical risk using a particular algorithm.**
  - **Swayamdipta et al. (2020) use the RoBERTa (Liu et al. 2019) model.**

- **The *<u>training dynamics of instance i</u>* are defined as statistics calculated across *E* epochs. The values of these measures serve as coordinates in our map.**

❖ Confidence, Correctness and Variability

➤ **Confidence** is the mean model probability of the true label (y_i^*) across epochs:

$$\hat{\mu}_i = \frac{1}{E}\sum_{e=1}^{E} p_{\theta^{(e)}}\left(y_i^* \mid x_i\right),$$

where $p_{\theta^{(e)}}$ denotes the model's probability with parameters $\theta^{(e)}$ at the end of the e^{th} epoch.

Remark:

Note that $\hat{\mu}_i$ is with respect to the true label y_i^*, not the probability assigned to the model's highest-scoring label.

- **Correctness** is the fraction of times the model correctly labels x_i across epochs.

  - **Remark**:

    Intuitively, a high-confidence instance is "easier" for the given learner.

- **Variability** measures the spread of $p_{\theta^{(e)}}\left(y_i^* \mid x_i\right)$ across epochs, using the standard deviation:

$$\hat{\sigma}_i = \sqrt{\frac{\sum_{e=1}^{E}\left(p_{\theta^{(e)}}\left(y_i^* \mid x_i\right) - \hat{\mu}_i\right)^2}{E}}$$

- **Remarks**:
  1. Variability also depends on the gold label y_i^*
  2. A given instance to which the model assigns the same label consistently (whether accurate or not) will have low variability; one which the model is indecisive about across training will have high variability.
  3. Variability, by definition, captures the uncertainty of the model.

Baselines

- **Swayamdipta et al. (2020) consider baselines using**
  - **all data**
  - **a 33% random sample**
  - **a mixture of *easy-to-learn* and *hard-to-learn* examples**
  - **They also consider baselines based on *high-and-low-correctness*.**

- **Remarks**:

1. **Training dynamics measures are used in data maps as coordinates for data points. They have intuitive connections with measures of *uncertainty*.**
2. **When a model fails to predict the correct label, the error may come from ambiguity inherent to the example (intrinsic uncertainty), or from the model's limitations (model uncertainty). It is important to separate these two sources of error in order to understand *how examples contribute to a dataset*.**

Conclusions:

- ✓ **Data maps** is an automatic method to visualize and diagnose large datasets using training dynamics.
- ✓ **Data maps** use effective simple training dynamics measures based on mean and standard deviation.
- ✓ **Data maps** can serve for constructing higher quality datasets and ultimately models that generalize better.
- ✓ **Data maps** represent an alternative to standard quantitative evaluation methods when comparing different model architectures trained on a given dataset.
- ✓ **Regional selections of data** improve model generalization and do so using less data, which potentially speeds up training.

- **The role of the three types of instances:**
  - ✓ ***ambiguous instances*** **are useful for high performance**
  - ✓ ***easy-to-learn*** **instances aid optimization**
  - ✓ ***hard-to-learn*** **instances often correspond to data errors.**

- **Note that selection of the optimal balance of *easy-to-learn* and *ambiguous* examples is an open problem.**

➢ An Application:

A Data Cartography – based MixUp for Pre-trained Language Models

- **MixUp (Zhang et al., 2018) is a *data augmentation strategy* where additional samples are generated during training by combining random pairs of training samples and their labels.**

- **The paper introducing MixUp:**

  Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. 2018. mixup: Beyond empirical risk minimization. In: Proceedings of the 6th International Conference on Learning Representations, ICLR 2018, Vancouver, Canada.

- **The idea behind standard MixUp:**

  In standard MixUp training data is augmented by linearly interpolating *random* training samples in the input space.

- **However:**
  - ✓ **Selecting *random pairs* in MixUp might not be optimal.**
  - ✓ **Park and Caragea (2022) focus on selecting *informative samples* for MixUp and propose TD-MixUp as a novel MixUp strategy.**

➢ **The studied problem: *random pairing* versus *informative pairing* for MixUp**

TD – MixUp

- TD – MixUp leverages training dynamics and allows more informative samples to be combined for generating new data samples.
- TD – MixUp first measures confidence, variability (Swayamdipta et al., 2020) and Area Under the Margin (AUM) (Pleiss et al., 2020) to identify the characteristics of training samples (e.g. as easy-to-learn or ambiguous samples), and then interpolates these characterized samples.

- **<u>The paper introducing TD – MixUp (2022)</u>:**

  Seo Yeon Park and Cornelia Caragea, A Data Cartography based MixUp for Pre-trained Language Models. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Juky 10-15, 2022, pp. 4244 – 4250.

- **Park and Caragea (2022) perform <u>interpolation</u> of <u>*easy-to-learn*</u> with <u>*ambiguous*</u> samples. Recall that:**
  - ***easy-to-learn* samples are useful for model optimization (parameter estimation)**
  - ***ambiguous* samples push the model to become more robust**

- **However:**

  the *easy-to-learn* and the *ambiguous* sets can contain mis-labeled samples that can degrade the model performance. Consequently, Park and Caragea (2022) measure another training dynamic, <u>Area Under the Margin (AUM)</u> introduced in (Pleiss et al., 2020).

- **AUM is used by Park and Caragea (2022) to filter out possibly mis-labeled samples in each set.**

Area Under the Margin (AUM)

- **The paper introducing AUM:**
  Geof Pleiss, Tianyi Zhang, Ethan R. Elenberg, and Kilian Q.Weinberger (2020). Identifying mislabeled data using the area under the margin ranking. In: Advances in Neural Information Processing Systems, volume 33.

- **AUM measures how different a true label for a sample is compared to a model's belief at each epoch.**

- **AUM is calculated as the average difference between the logit values for a sample's assigned class (gold label) and its highest non-assigned class across training epochs.**

- **Formally:**
  - **given a sample (x_i, y_i), we compute $AUM\ (x_i, y_i)$ as the area under the margin averaged across all training epochs E.**
  - **Specifically, at some epoch $e \in E$, the margin is defined as:**

    $$M^e(x_i, y_i) = z_{y_i} - max_{y_i != k}(z_k)$$

    where:

    - $M^e(x_i, y_i)$ is the margin of sample x_i with the true label y_i,

    - z_{y_i} is the logit corresponding to the true label y_i,

    - $max_{y_i != k}(z_k)$ is the largest <u>*other*</u> logit corresponding to label k not equal to y_i.
  - **The AUM of (x_i, y_i) across all epochs is:**

$$AUM\ (x_i, y_i) = \frac{1}{E} \sum_{e=1}^{E} M^e\ (x_i, y_i)$$

Confidence versus AUM:

Both **AUM** and **confidence** measure training dynamics, but:

- **confidence** simply measures the probability output of the gold label and how much it fluctuates over the training epochs
- **AUM** measures the probability output of the gold label with respect to the model's belief in what the label for a sample should be according to its generalization capability (derived by observing other similar samples during training).

- **Identifying possibly mis-labeled data in**
  - **the set of *easy-to-learn* samples**
  - **the set of *ambiguous* samples**

- **Steps:**
  1. **Fine-tune a model on each set, respectively, with inserting fake data (i.e. *threshold samples*)**
     - **Constrcting *threshold samples*:**

       Take a subset of the training data and re-assign their labels randomly, including a class that does not really exist.
  2. **Data with similar or worse AUMs than threshold samples can be assumed to be mis-labeled (Pleiss et al., 2020)**

Specifically (identifying mis-labeled data):

- **Given N training samples that belong to c classes, randomly select $\frac{N}{c+1}$ samples per class and re-assign their labels to classes that are different from the original class.**
- **Train a model on training samples including threshold samples and measure the AUMs of all training data.**
- **Identify possible mis-labeled data by computing a threshold value (i.e. the k-th percentile threshold sample AUMs where k is a hyper-parameter chosen on the validation set).**
- **Filter out samples that have lower AUM than the threshold value.**

MixUp

MixUp training generates vicinity training samples according to the rule introduced by Zhang et al. (2018):

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

where:

x_i **and** x_j **are two randomly sampled input points**

y_i **and** y_j **are their associated one-hot encoded labels**

λ **is a mixing ratio sampled from a** Beta(α, α) **distribution with a hyper-parameter** α.

- **Park and Caragea (2022) propose TD – MixUp, a novel MixUp strategy that leverages Training Dynamics and allows more informative samples to be combined for generating new data samples.**

- **Note that:**
  - **In standard MixUp training data is augmented by linearly interpolating *random* training samples in the input space.**
  - **In TD – MixUp one *easy-to-learn* sample is interpolated with one *ambiguous* sample after applying AUM to filter potential erroneous samples that harm performance.**

Experiments and Results (see the paper for details)

- **Tasks**

  TD – MixUp is evaluated (Park and Caragea, 2022) on 3 natural language understanding tasks:

  - **Natural Language Inference (NLI)**: to predict if the relation between a hypothesis and a premise is entailment, contradiction or neutral. Multi-Genre Natural Language Inference (MNLI) captures NLI with diverse domains.
  - **Paraphrase Detection**: testing if two questions are semantically equivalent
  - **Commonsense Reasoning**: Situations with Adversarial Generations (SWAG) is a commonsense reasoning task to choose the most plausible continuation of a sentence among four candidates.

Experimental setup

- **Metrics:**
  - ✓ ***accuracy*** **– should be high**
  - ✓ ***expected calibration error (ECE)*** **(Guo et al., 2017; Desai and Durrett, 2020) – should be low**
    - **Note that:**

Expected Calibration Error (ECE) measures how well a model's estimated "probabilities" match the true (observed) probabilities by taking a weighted average over the absolute difference between accuracy and confidence.

Experimental Setup

- **Park and Caragea (2022) use BERT and follow the published train/validation/test datasets splits as described in Desai and Durrett (2020).**

- **Park and Caragea (2022) report results averaged across 5 fine-tuning runs with random restarts for all experiments.**

Baseline Methods

- **BERT** (Devlin et al., 2019)
- **Back Translation Data Augmentation** (Edunov et al., 2018)
- **MixUp** (Zhang et al., 2018) – generates augmented samples by interpolating random training samples in the input space (obtained from the first layer of the BERT pre-trained language model)
- **Manifold MixUp** (M-MixUp) (Verma et al., 2019) – generates additional samples by interpolating random training samples in the feature space (obtained from the task-specific layer on top of the BERT pre-trained language model).
- **MixUp for Calibration** (Kong et al., 2020) – generates augmented samples by utilizing the cosine distance between samples in the feature space.

- **Remark**

  All previously mentioned baselines use 100% training data. The method proposed by Park and Caragea (2022) focuses on particular subsets of the training data.

Technique

To explore the effect of different subsets of the data that are characterized using training dynamics, Park and Caragea (2022) compare the result of BERT fine-tuned on 100% training data with the results of BERT when fine-tuned on these subsets.

- **<u>Remarks</u>:**

1. **The focus is on particular subsets of the training data.**
2. **Accuracy and ECE improve when filtering out possibly mislabeled samples in the top 33% *easy-to-learn* samples by using AUM in all cases. Specifically, using 24% train, *easy-to-learn* with AUM returns better accuracy and lower ECE than 33% train, *easy-to-learn*. This shows that there are some potentially erroneous samples that harm the performance in the top 33% *easy-to-learn* samples.**
3. **<u>In contrast</u>: in many cases accuracy and ECE worsen when we filter out possibly mislabeled samples in the 33% *ambiguous* samples by using AUM. This suggests that <u>all *ambiguous* samples are useful for learning and generalization</u>.**
4. **Fine-tuning BERT on both the *easy-to-learn* and the *ambiguous* samples (66% train, *easy-to-learn* and *ambiguous*) achieves similar performance and ECE as 100% train.**

➢ Ablation Study

- **Comparing the impact of the MixUp operation on samples generated by *random pairing* versus samples generated by *informative pairing*:**

  Park and Caragea (2022) compare the results of MixUp on 66% train set (MixUp operation between randomly selected samples on 66% train set, which is the union of the top 33% *easy-to-learn* and the top 33% *ambiguous* samples) and the proposed TD-MixUp (MixUp operation conducted between the *easy-to-learn* filtered by AUM and the *ambiguous* samples).

- **Conclusion of the ablation study:**

TD-MixUp (which selects *informative samples* to combine) performs better with respect to accuracy and ECE than vanilla MixUp (that selects *random samples*) in all cases [see tables with numerical results in (Park and Caragea, 2022)].

❖ General conclusion

- **Generating augmented samples:**

  When studying random pairing versus informative pairing, the conclusion is that we should allow more informative samples to be combined in generating augmented samples.

  - ✓ **Specifically, we should use the _easy-to-learn samples_ filtered by AUM together with _all the ambiguous samples_.**